

1. Administrative & Study Identification

Full Study Title: A Phase III, Multicenter, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Efficacy and Safety Study of Trontinemab in Participants With Early Symptomatic Alzheimer's Disease (MCI to Mild Dementia Due to AD) **Short Title/ Acronym:** TRONTIER 1 **Protocol Number:** WN45443 **NCT Number:** NCT07169578 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Trontinemab **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of trontinemab in participants with early symptomatic Alzheimer's disease (AD) (mild cognitive impairment [MCI] to mild dementia due to AD), measured by the change from baseline to Week 72 in the Clinical Dementia Rating, Sum of Boxes (CDR-SB). **Secondary Objectives:** To evaluate changes from baseline through Week 72 in cognitive and functional abilities using the Alzheimer's Disease Assessment Scale-Cognition 13 (ADAS-Cog-13), Alzheimer's Disease Cooperative Study Activities of Daily Living Inventory (ADCS-ADL), Integrated Alzheimer's Disease Rating Scale (iADRS), and Mini-Mental State Examination (MMSE).

2.2 Background & Rationale To address the progressive cognitive and functional decline in patients with early symptomatic Alzheimer's Disease. Alzheimer's disease is characterized by the accumulation of beta-amyloid plaques and tau tangles. This study evaluates whether trontinemab, an experimental medication designed to remove amyloid plaques from the brain, can effectively modify disease progression and preserve cognitive function when administered during the early stages of cognitive decline.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Quadruple blind: Participant, Care Provider, Investigator, Outcomes Assessor) **Randomization:** Randomized, Parallel-Group (1:1 ratio)

3.2 Planned Enrollment & Duration Target \$N\$: 800 participants **Number of Sites:** 134 locations globally **Estimated Study Start:** 17-Sep-2025 **Estimated Primary Completion:** 07-Jun-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 50 to 90 years old. 2. Diagnosis of probable AD dementia or MCI due to AD (screening MMSE score ≥ 22 and CDR-GS of 0.5 or 1.0). 3. Evidence of abnormal AD pathological process (amyloid accumulation) confirmed via amyloid PET scan or cerebrospinal fluid (CSF) testing. 4. Availability of a study partner willing to support and participate in the research.

4.2 Exclusion Criteria 1. Any evidence of a condition other than AD that may affect cognition. 2. History or presence of clinically significant cerebrovascular disease, CNS trauma, or intracranial mass. 3. MRI evidence of significant cerebral abnormalities, inability to tolerate MRI procedures, or contraindication to MRI. 4. Any other unstable or inadequately controlled medical conditions (e.g., cardiovascular, hepatic, renal disease) that could interfere with safety assessments.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Trontinemab or matching Placebo administered at scheduled intervals. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** 72 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Stable medications for adequately controlled co-morbidities; eyewear and hearing aids are permitted for cognitive testing. **Prohibited:** Investigational medications outside this study; medications for non-stable medical conditions that could interfere with assessments or participant safety.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, MRI, Amyloid PET scan/CSF test, MMSE, CDR-GS
Baseline	Day 0	Randomization, First IV Infusion
Interim	Scheduled	IV Infusions, Safety Labs, Cognitive/Functional Assessments (ADAS-Cog-13, ADCS-ADL)
End of Study	Week 72	Final Efficacy Assessments (CDR-SB, iADRS), Final Safety Review

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline to Week 72 in the Clinical Dementia Rating, Sum of Boxes (CDR-SB). **Measurement Method:** Standardized Clinical Dementia Rating assessment scale administered by a blinded outcomes assessor.

7.2 Secondary & Exploratory Endpoints * Change from baseline through Week 72 in Alzheimer's Disease Assessment Scale-Cognition 13 (ADAS-Cog-13). * Change from baseline through Week 72 in Alzheimer's Disease Cooperative Study Activities of Daily Living Inventory (ADCS-ADL) total score and instrumental score. * Change from baseline through Week 72 in Integrated Alzheimer's Disease Rating Scale (iADRS) and MMSE.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of emergent adverse events, laboratory tests, and neurological examinations. **Serious Adverse Events (SAEs):** Reported per standard regulatory protocols for Phase III trials. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee is in place to review safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy, Safety Set for safety parameters. **Sample Size Justification:** \$N=800\$ appropriately powered to detect statistically significant differences in CDR-SB scores between the trontinemab and placebo groups over 72 weeks. **Handling of Missing Data:** Standard imputation methods for longitudinal cognitive testing drop-outs.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard onsite monitoring for IV administration safety, cognitive assessment fidelity, and biomarker tracking. **Audit Trail:** eCRF locking and tracking of IV infusion adherence, ensuring quadruple-blinding integrity.

11. Study Identifiers & Governance

Primary ID: WN45443 **Registry Number:** NCT07169578 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** TRONTIER 1 **Official Regulatory Title:** A Phase III, Multicenter, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Efficacy and Safety Study of Trontinemab in Participants With Early Symptomatic Alzheimer's Disease (MCI to Mild Dementia Due to AD)

12. Clinical Framework (Alzheimer's Disease Specific)

Primary Condition: Early Symptomatic Alzheimer's Disease (MCI to Mild Dementia) **Diagnosis Threshold:** MMSE ≥ 22 , CDR-GS ≥ 0.5 or ≥ 1.0 , and positive amyloid PET/CSF **Medical Methodology:** Investigational monoclonal antibody therapy targeting beta-amyloid **Optimization Target:** Slowing the progression of cognitive and functional decline

13. Study Procedures & Methodology

Management Pathway: Amyloid plaque clearance via Intravenous (IV) Infusion **Education Protocol (HCP):** Blinded administration and standardized neurocognitive rating training **Education Protocol (Patient):** Mandatory inclusion of a dedicated study partner to support participation and assist with daily living assessments **Quality Audit Mechanism:** Centralized scoring review for CDR-SB and independent MRI/PET scan reads

14. Observation & Follow-Up Schedule

Total Duration: 72 Weeks **Onsite Visit Cadence:** Regular visits for IV infusions and clinical assessments **Remote Monitoring Frequency:** As specified by clinical site guidelines for AE follow-up **Checklist Review Interval:** Cognitive and functional scale reviews spanning up to Week 72

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				